GPU compatibility from the TensorFlow Installation Guide

- URL: https://www.tensorflow.org/install/gpu?hl=es-419
- Software requirements:
    - NVIDIA GPU drivers: https://www.nvidia.com/download/index.aspx?lang=en-us
    - CUDA Toolkit: https://developer.nvidia.com/cuda-toolkit-archive
    - CUPTI: http://docs.nvidia.com/cuda/cupti/
    - cuDNN SDK: https://developer.nvidia.com/cudnn
- Check what is installed running "nvidia-smi"

Using Windows Subsystem for Linux (WSL)

- WSL is a feature of Windows that allows you to run a Linux environment on your Windows machine.
- On Windows it is a recommended option to use the GPU libraries.
- Install WSL: https://learn.microsoft.com/es-es/windows/wsl/install
    - Powershell => "`wsl --install`"
- Restart => The Linux console will start asking for a user and password.
- To open the Linux console search for Ubuntu.
- To check in the console that the drivers are installed type "nvidia-smi".
- We install tensorflow
    - o Note (if problems with all of the following)
        - Look DNS => `sudo nano /etc/resolv.conf`
        - Include "`nameserver 8.8.8.8`" y "`nameserver 8.8.4.4`"
    - First install pip
    - We do "`sudo apt-get update`" to update the base of packages.
    - We do "`sudo apt install python3-pip`" to install pip
    - We install tensorflow "`pip install tensorflow[and-cuda]`"
    - o Check the CPU
        - `python3 -c "import tensorflow as tf; print(tf.reduce_sum(tf.random.normal([1000, 1000])))"`
        - At the end it has to say something like "`tf.Tensor(202.3931, shape=(), dtype=float32)`"
    - And the GPU
        - `python3 -c "import tensorflow as tf; print(tf.config.list_physical_devices('GPU'))"`
        - At the end it has to say something like "`[PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]`"
    - o We install the libraries as if we were in an environment
        - `keras, notebook, matplotlib`
    - WSLs home de WSL is `/home/username`
        - The Windows folder is at `/mnt/c`
    - Run notebook
        - `python3 -m notebook`
        - Copy the local address of the notebook and open it in the browser